

SageMath: A Second Glance at Cybersecurity

The eighth article in the series on SageMath explores a classical encryption scheme called the Rail Fence cipher and introduces the concept of symmetric-key encryption.



In the previous article in this SageMath series (published in the November 2024 issue of *OSFY*), we began exploring cybersecurity and introduced classical encryption techniques, setting a foundation for further discussion. In this article, we delve into a classical encryption scheme known as the Rail Fence cipher. Unlike the Caesar and Vigenère ciphers we covered earlier—both of which are substitution ciphers that replace each plaintext letter with another to create the ciphertext—the Rail Fence cipher is a transposition cipher. Here, letters in the plaintext are rearranged rather than substituted, producing a scrambled version of the plaintext to form the ciphertext. Notice that any terms introduced without explanation here were discussed in the previous article.

Before proceeding further, we need to focus on two additional aspects as we progress in this series, beyond discussing SageMath code. First, we need to set up effective tools for working with SageMath. Second, we should explore the historical context of our topic: cybersecurity and cryptography.

Let us start by addressing a tool-related issue. So far, we have been using an online platform called CoCalc, a web-based tool well-suited for running SageMath programs. However, CoCalc operates on a freemium model, requiring payment for certain features. While I am not an ardent advocate of exclusively using free and open source software, I initially recommended CoCalc for its accessibility. Since SageMath has a steep learning curve, I thought it would be